

Lab 1 - Check My Grade Application

Approach

For this lab, I designed the CheckMyGrade application as a console based system that servers professors and students, each with their own tailored menu interface. The lab instructions did not explicitly specify whether the application should be purely backend or user-facing, so I made all functionally accessible through an interactive console interface. Rather than creating a separate administrator role, the professor role was extended to handle operations that would typically require administrative access. A special rank of “Professor & Administrator” was coded to grant privileges to perform backend level operations such as adding, deleting, and modifying student records, courses, and professor accounts. All other professors retain standard access such as viewing their course, managing grades, and searching student records. Additionally, backend operations such as bulk data manipulation and performance testing are handled through the unit test script.

Data files

- All student, course, and professor data is persisted across sessions through four CSV files: **students.csv, courses.csv, professor.csv, and login_encrypted.csv**
- Additions, deletions, and modifications made during a session are immediately reflected and written back to the corresponding file.

Note: The original CSV files were randomly generated to test the code while building the application. To protect this data during testing, I kept the files produced from unit tests separate from original files. Before each test, all real CSV files are backed up, and after each test they are restored. The unit tests write exclusively to separate test files: **test_students.csv, test_course.csv, test_professors.csv**, which will be submitted alongside the original files.

Object Oriented Design

- Related data and behavior are stored into distinct classes. For example, Student and Professor inherit shared attributes from a base Person class (IS-A), and service classes (StudentService, ProfessorService, CourseService) compose domain objects to manage all application logic (HAS-A). Please refer to the class diagram attached below.

Data structures included

- Python **lists** are used as the primary array-based data structure throughout the application for storing all student, professor, course, and user records. Records are loaded from CSV into lists and operations are performed directly on lists before being written back to CSV

```
def load_courses(filename):
    """Load courses as dict: {course_id: {course_name, credits}}"""
    courses = {}
    with open(filename, newline='') as file:
        reader = csv.DictReader(file)
        for row in reader:
            courses[row["course_id"]] = {"course_name": row["course_name"],
                                         "credits": int(row["credits"])}
    return courses
```

```
def search_by_name(self, name):
    name_lower = name.lower()
    return [s for s in self.students
            if name_lower in s.first_name.lower()
            or name_lower in s.last_name.lower()]
```

- Stack** is implemented using LIFO logic, and stores grade history when changes are made to grade. It allows professors to reverse their last grade change if needed in `add_grade()` and `delete_grade()`

```
class Stack:
    """LIFO structure for undo grade history."""
    def __init__(self):
        self.items = []

    def push(self, item):
        self.items.append(item)

    def pop(self):
        if self.is_empty():
            return None
        return self.items.pop()

    def peek(self):
        if self.is_empty():
            return None
        return self.items[-1]

    def is_empty(self):
        return len(self.items) == 0

    def size(self):
        return len(self.items)
```

```
self.grade_undo_stack.push({
    "student_id": found.student_id,
    "course_id": found.course_id,
    "grade": found.grade,
    "mark": found.mark
})
found.grade = new_grade
found.mark = new_mark
```

- **Queue** is implemented to manage student enrollment in batches in `add_new_student()` using FIFO processing. This enrolls students in the order they were entered.

```
class Queue:
    """FIFO structure for student enrollment processing."""
    def __init__(self):
        self.items = []

    def enqueue(self, item):
        self.items.append(item)

    def dequeue(self):
        if self.is_empty():
            return None
        return self.items.pop(0)

    def is_empty(self):
        return len(self.items) == 0

    def size(self):
        return len(self.items)
```

```
new_student = Student(
    student_id, email, first_name, last_name,
    prof.course_id, "", 0
)
enrollment_queue.enqueue(new_student)
print(f"Added {first_name} {last_name} to enrollment queue.")

if enrollment_queue.is_empty():
    print("No students to enroll.")
    return

count = 0
while not enrollment_queue.is_empty():
    student = enrollment_queue.dequeue()
```

- **Hash Table** is implemented to provide $O(1)$ course lookup by course ID, and is used in ProfessorService via `get_course_info()` which is called by `professor_details()`, `display_grade_report()`, and `show_course_details_by_professor()`, and in StudentService via `self.course_index.lookup()` inside `display_records()`, `display_grade_report()`, `display_marks_only()`, and `display_grades_only()`.

```
class HashTable:
    """Hash table for fast O(1) course lookup by course_id."""
    def __init__(self):
        self.table = {}

    def insert(self, key, value):
        """Insert a course into the hash table."""
        self.table[key] = value

    def lookup(self, key):
        """Look up a course by course_id - O(1)."""
        return self.table.get(key, None)

    def delete(self, key):
        """Remove a course from the hash table."""
        if key in self.table:
            del self.table[key]

    def build_from_dict(self, courses_dict):
        """Build hash table from courses dict."""
        for course_id, info in courses_dict.items():
            self.table[course_id] = info

    def size(self):
        return len(self.table)
```

```
print(f"\nEnrolled Courses:")
print(f"{'Course ID':<12} {'Course Name':<30} {'Credits':<8}")
print("-" * 52)
for s in records:
    info = self.course_index.lookup(s.course_id)
    course_name = info["course_name"] if info else "N/A"
    course_credits = info["credits"] if info else "N/A"
    print(f"{s.course_id:<12} {course_name:<30} {course_credits:<8}")
```

Password Encryption/ Decryption

- Password security is handled through a Caesar cipher implemented in the TextSecurity class, which has both encrypt_password() and decrypt_password() methods. Passwords are stored in encrypted form in login_encrypted.csv and decrypted at login for verification.

```
(base) amy@amy3-air-2: /usr/local/bin/python3 -- /usr3/amy/amy/desktop/data-200 -- Python/Ed-1/Run_Ed-1
Welcome to CheckMyGrade! This is an evaluation tool for assessing students' grades.
There are different access areas depending on your role.
Please login to proceed...
User Id (email): robert.parker10@university.edu
Password: 4eAD161

Password encrypted: 8iEH505
Password decrypted: 4eAD161
Match verified: True
Login successful. Role: Professor & Administrator
**** Professor Menu ****
1. View my profile
2. Display course grade report
3. Manage student grade(s)
4. View course details
5. Add student(s) to my course
6. Remove student from my course
7. Manage courses
8. Manage professors
9. Change password
10. Logout
What would you like to do? Choose one of the above options: 9
```

- When a user changes their password, it is encrypted and stored in login_encrypted.csv. When the user logs in the next time, their new password will go through successfully.

```
10. Logout
What would you like to do? Choose one of the above options: 9
Enter your current password: 4eAD161
Enter NEW password: newPassword232
Confirm NEW password: newPassword232
Password updated successfully!
```

```
User Id (email): robert.parker10@university.edu
Password: newPassword232

Password encrypted: riaTewasvh676
Password decrypted: newPassword232
Match verified: True
Login successful. Role: Professor & Administrator
```

Note: Debugging password is shown to demonstrate encryption/ decryption works. It will be commented out in the code when submitted.

Add/Delete/Modify Professor records

- The Professor with the “Professor & Administrator” role is allowed to perform changes to professor records (Menu item #8)
- Adding, deleting, or modifying professor information will update **professor.csv** with the changes and print a confirmation message to the console, and display the refreshed data as the menu loops back to the top.

```
What would you like to do? Choose one of the above options: 8

===== Professor List =====
ID      Name      Rank      Course
-----
P001    Laura Clark  Senior Professor  CSE101A
P002    Sarah Turner  Assistant         CSE101A
P003    Sarah Morris  Assistant Professor  MAT202B
P004    Anna Evans   Associate Professor  MAT202B
P005    Laura Rogers  Instructor         ENG150C
P006    Emily Parker  Assistant Professor  PHY210D
P007    Jessica Morris Senior Professor    PHY210D
P008    James Reed   Associate Professor  BI0110E
P009    Michael Reed Instructor       HIS330F
P010    Robert Parker Professor & Administrator  HIS330F
P011    Linda Adams  Assistant Professor  CHE115G
P012    Linda Campbell Instructor       CIS220H
P013    Karen Adams  Instructor       ART105I
=====

1. Add a professor
2. Delete a professor
3. Modify professor details
4. Back
Choose an option: 1
Generated Professor ID: P014
First name: Linda
Last name: Jones
Email: linda.jones@sjsu.edu
Course ID: DATA226
Rank: Senior Professor
Professor added successfully.

===== Professor List =====
ID      Name      Rank      Course
-----
P001    Laura Clark  Senior Professor  CSE101A
P002    Sarah Turner  Assistant         CSE101A
P003    Sarah Morris  Assistant Professor  MAT202B
P004    Anna Evans   Associate Professor  MAT202B
P005    Laura Rogers  Instructor         ENG150C
P006    Emily Parker  Assistant Professor  PHY210D
P007    Jessica Morris Senior Professor    PHY210D
P008    James Reed   Associate Professor  BI0110E
P009    Michael Reed Instructor       HIS330F
P010    Robert Parker Professor & Administrator  HIS330F
P011    Linda Adams  Assistant Professor  CHE115G
P012    Linda Campbell Instructor       CIS220H
P013    Karen Adams  Instructor       ART105I
P014    Linda Jones   Senior Professor    DATA226
=====
```

Add/Delete/Modify Professor records cont...

```
1. Add a professor
2. Delete a professor
3. Modify professor details
4. Back
Choose an option: 2
Enter professor ID to delete: P002
Professor deleted successfully.
```

```
===== Professor List =====
ID      Name      Rank      Course
-----
P001    Laura Clark   Senior Professor   CSE101A
P003    Sarah Morris  Assistant Professor MAT202B
P004    Anna Evans    Associate Professor MAT202B
P005    Laura Rogers  Instructor         ENG150C
P006    Emily Parker  Assistant Professor PHY210D
P007    Jessica Morris Senior Professor   PHY210D
P008    James Reed    Associate Professor BI0110E
P009    Michael Reed  Instructor         HIS330F
P010    Robert Parker Professor & Administrator HIS330F
P011    Linda Adams   Assistant Professor CHE115G
P012    Linda Campbell Instructor        CIS220H
P013    Karen Adams   Instructor        ART105I
P014    Linda Jones   Senior Professor   DATA226
=====
```

```
1. Add a professor
2. Delete a professor
3. Modify professor details
4. Back
Choose an option: 3
Enter professor ID to modify: P012
Leave blank to keep current value.
New rank: Associate Professor
New course ID: CIS220H
Professor details updated successfully.
```

```
===== Professor List =====
ID      Name      Rank      Course
-----
P001    Laura Clark   Senior Professor   CSE101A
P003    Sarah Morris  Assistant Professor MAT202B
P004    Anna Evans    Associate Professor MAT202B
P005    Laura Rogers  Instructor         ENG150C
P006    Emily Parker  Assistant Professor PHY210D
P007    Jessica Morris Senior Professor   PHY210D
P008    James Reed    Associate Professor BI0110E
P009    Michael Reed  Instructor         HIS330F
P010    Robert Parker Professor & Administrator HIS330F
P011    Linda Adams   Assistant Professor CHE115G
P012    Linda Campbell Associate Professor CIS220H
P013    Karen Adams   Instructor        ART105I
P014    Linda Jones   Senior Professor   DATA226
=====
```

Add/Delete/Modify Courses

- The Professor with the “Professor & Administrator” role is allowed to perform changes to course records
- Adding, deleting, or modifying course information will update course.csv with the changes, print a confirmation message to the console, and display the refreshed data as the menu loops back to the top.

```
10. Logout
What would you like to do? Choose one of the above options: 7

===== Course List =====
Course ID    Course Name    Credits
-----
ART105I      Art Appreciation    2
BI0110E      Biology Fundamentals  3
CHE115G      General Chemistry    3
CIS220H      Data Structures      4
CSE101A      Introduction to Programming  6
ENG150C      English Composition    3
HIS330F      World History        3
MAT202B      Calculus II          4
PHY210D      Intro to Physics      3

1. Add a course
2. Delete a course
3. Modify a course
4. Back
Choose an option: 2
Enter course ID to delete: PHY210D
Warning: 2 professor(s) are currently assigned to PHY210D:
  P006 - Emily Parker
  P007 - Jessica Morris
Deleting this course will set their course assignment to 'UNASSIGNED'.
Are you sure? Type 'yes' to confirm: yes
Course deleted successfully.
2 professor(s) updated to UNASSIGNED.

===== Course List =====
Course ID    Course Name    Credits
-----
ART105I      Art Appreciation    2
BI0110E      Biology Fundamentals  3
CHE115G      General Chemistry    3
CIS220H      Data Structures      4
CSE101A      Introduction to Programming  6
ENG150C      English Composition    3
HIS330F      World History        3
MAT202B      Calculus II          4
```

Add/Delete/Modify Courses cont...

```
10. Logout
What would you like to do? Choose one of the above options: 7

===== Course List =====
Course ID   Course Name           Credits
-----
ART105I     Art Appreciation      2
BI0110E     Biology Fundamentals   3
CHE115G     General Chemistry      3
CIS220H     Data Structures        4
CSE101A     Introduction to Programming 6
ENG150C     English Composition     3
HIS330F     World History          3
MAT202B     Calculus II            4
PHY210D     Intro to Physics       3

1. Add a course
2. Delete a course
3. Modify a course
4. Back
Choose an option: 1
Enter new course ID: DATA226
Enter course name: Data Warehouse
Enter credits (1-6): 3
Course added successfully.

===== Course List =====
Course ID   Course Name           Credits
-----
ART105I     Art Appreciation      2
BI0110E     Biology Fundamentals   3
CHE115G     General Chemistry      3
CIS220H     Data Structures        4
CSE101A     Introduction to Programming 6
ENG150C     English Composition     3
HIS330F     World History          3
MAT202B     Calculus II            4
PHY210D     Intro to Physics       3
DATA226     Data Warehouse         3

1. Add a course
2. Delete a course
3. Modify a course
4. Back
Choose an option: 3
Enter course ID to modify: DATA226
Leave blank to keep current value.
New course name:
New credits (1-6): 4
Course updated successfully.
```

```
DATA226     Data Warehouse         4

1. Add a course
2. Delete a course
3. Modify a course
4. Back
Choose an option: 3
Enter course ID to modify: DATA226
Leave blank to keep current value.
New course name:
New credits (1-6): 4
Course updated successfully.

===== Course List =====
Course ID   Course Name           Credits
-----
ART105I     Art Appreciation      2
BI0110E     Biology Fundamentals   3
CHE115G     General Chemistry      3
CIS220H     Data Structures        4
CSE101A     Introduction to Programming 6
ENG150C     English Composition     3
HIS330F     World History          3
MAT202B     Calculus II            4
DATA226     Data Warehouse         4
```


Adding/Deleting/modify student records

- Professors can make grade updates for students enrolled in their class which will update in **student.csv**

```
1 result(s) found in 0.000050 seconds.
**** Professor Menu ****
1. View my profile
2. Display course grade report
3. Manage student grade(s)
4. View course details
5. Add student(s) to my course
6. Remove student from my course
7. Manage courses
8. Manage professors
9. Change password
10. Logout
What would you like to do? Choose one of the above options: 3
1. Add/Update Grade
2. Delete grade
3. Return
Which action would you like to perform?: 1
Enter student Id: S087
Making updates to Harper Jones's grade for HIS330F
Current grade: F (50)
Updating existing grade...
Enter grade(A-F): D
Enter mark(0-100): 68
Student grade updated successfully!
Would you like to undo this change? (yes/no): no
1. Add/Update Grade
2. Delete grade
3. Return
Which action would you like to perform?: 2
Enter student Id: S087
Making updates to Harper Jones's grade for HIS330F
Current grade: D (68)
Grade removed.
Would you like to undo this change? (yes/no): yes
Undone. Harper Jones's grade restored to D (68).
1. Add/Update Grade
2. Delete grade
3. Return
Which action would you like to perform?: 3
```

- Professors can also add/delete new students to their course. Addition is reflected in students.csv, which updates in course details when we print it again

Here, we see that Jane Doe is added, and Amy Tan is removed

```
10. Logout
What would you like to do? Choose one of the above options: 5

Enter student details (or press Enter on first name to finish):
First name (or press Enter to finish): Jane
Last name: Doe
Email: janedoe@uni.com
Generated Student ID: S096
Added Jane Doe to enrollment queue.

Enter student details (or press Enter on first name to finish):
First name (or press Enter to finish):

1 student(s) successfully enrolled in HIS330F.
```

```
**** Professor Menu ****
What would you like to do? Choose one of the above options: 6
Enter student ID to remove (or press enter to cancel): S093
Are you sure you want to remove Amy Tan from HIS330F?
Type 'yes' to confirm: yes
Amy Tan removed from HIS330F.
**** Professor Menu ****
1. View my profile
2. Display course grade report
3. Manage student grade(s)
4. View course details
5. Add student(s) to my course
6. Remove student from my course
7. Manage courses
8. Manage professors
9. Change password
10. Logout
What would you like to do? Choose one of the above options: 4
```

Adding/Deleting/modify student records cont..

After a student is removed/added to a course, when we print Course Details after the updates, the updates to student records will be reflected:

```
What would you like to do? Choose one of the above options: 4
===== Course Details =====
Course ID:   HIS330F
Course name: World History
Credits:     3
Taught by:
  Professor Michael Reed (Instructor)
  Professor Robert Parker (Professor & Administrator)
=====
Students enrolled:
Student Id   Name
-----
S001         UpdatedName Jones
S003         Mia Davis
S004         Ethan Williams
S009         Lucas Garcia
S010         Benjamin Davis
S016         Evelyn Martin
S017         Isabella Johnson
S023         Ethan Garcia
S027         Ava Brown
S031         Ava Jackson
S032         Logan Jackson
S033         Olivia Martin
S036         Mia Jones
S039         Abigail Miller
S044         Sophia Hernandez
S046         James Taylor
S056         Ava Johnson
S057         Benjamin Gonzalez
S064         Isabella Smith
S067         Isabella Moore
S070         Harper Martinez
S071         Lucas Taylor
S074         Liam Williams
S075         Evelyn Martin
S077         James Moore
S078         Olivia Miller
S082         Abigail Lopez
S083         Harper Garcia
S085         Mia Johnson
S086         Isabella Anderson
S087         Harper Jones
S089         Isabella Williams
S091         Emma Davis
S092         Isabella Anderson
S096         Jane Doe
=====
Total students enrolled: 35
```

- Students can edit their personal information such as first/last name, which will be updated in students.csv

```
7. Logout
What would you like to do? Choose one of the above options: 3
===== Student Profile =====
Student ID: S003
Name:      Mia Davis
Email:     mia.davis03@university.edu
=====

Enrolled Courses:
Course ID   Course Name           Credits
-----
HIS330F     World History          3
CHE115G     General Chemistry      3
CSE101A     Introduction to Programming 6
PHY210D     N/A                    N/A
MAT202B     Calculus II            4

Would you like to edit any information? (yes/no): yes
1. Update first name
2. Update last name
3. Back
Choose an option: 2
Enter new last name: Davison
Profile updated successfully.
***** Student Menu *****
1. Check my grades
2. Check my marks
3. View my profile
4. View grade report
5. View course catalog
6. Change password
7. Logout
What would you like to do? Choose one of the above options: 3
===== Student Profile =====
Student ID: S003
Name:      Mia Davison
Email:     mia.davis03@university.edu
=====

Enrolled Courses:
Course ID   Course Name           Credits
-----
HIS330F     World History          3
CHE115G     General Chemistry      3
```

- Professors can view their profile, which contains their personal info, and related courses.

```

Login successful. Role: Professor & Administrator
***** Professor Menu *****
1. View my profile
2. Display course grade report
3. Manage student grade(s)
4. View course details
5. Add student(s) to my course
6. Remove student from my course
7. Manage courses
8. Manage professors
9. Change password
10. Logout
What would you like to do? Choose one of the above options: 1
===== Professor Details =====
Name: Robert Parker
Professor ID: P010
Email: robert.parker10@university.edu
Rank: Professor & Administrator
Course taught: HIS330F - World History (3 credits)
=====

```

Generate grade reports categorized by course & professor

- Includes **statistical functions**: average mark, median mark

```

10. Logout
What would you like to do? Choose one of the above options: 2

Course grade report for HIS330F: World History (3 credits)
Student Id  Name                Grade  Marks
=====
S001        UpdatedName Jones      D      64
S003        Mia Davis              A      98
S004        Ethan Williams         C      70
S009        Lucas Garcia           A      99
S010        Benjamin Davis         A      98
S016        Evelyn Martin          D      65
S017        Isabella Johnson       C      78
S023        Ethan Garcia           F      53
S027        Ava Brown              C      76
S031        Ava Jackson            B      86
S032        Logan Jackson          C      75
S033        Olivia Martin          C      78
S036        Mia Jones              D      63
S039        Abigail Miller         B      83
S044        Sophia Hernandez       A      98
S046        James Taylor           A      94
S056        Ava Johnson            A      93
S057        Benjamin Gonzalez     D      62
S064        Isabella Smith         C      79
S067        Isabella Moore         A      93
S070        Harper Martinez        F      57
S071        Lucas Taylor           B      82
S074        Liam Williams          B      89
S075        Evelyn Martin          D      67
S077        James Moore            C      74
S078        Olivia Miller          A      96
S082        Abigail Lopez          C      78
S083        Harper Garcia          D      62
S085        Mia Johnson            A      95
S086        Isabella Anderson      D      61
S087        Harper Jones           F      50
S089        Isabella Williams      A      95
S091        Emma Davis             A      91
S092        Isabella Anderson      B      87
=====

Average mark: 79.09
Median mark: 78.50

Grade Distribution
Grade  Count  Percentage  Marks Range
=====
A      11      32.4%      90-100
B       5      14.7%      80-89
C       8      23.5%      70-79
D       7      20.6%      60-69
F       3       8.8%       0-59
=====

1. Search students
2. Back
Choose an option: 1

```

Sorting/Searching through records

- Includes search functionality for professor to search student records
- prints time taken to search the records

```
6. Remove student from my course
7. Search student records
8. Change password
9. Logout
What would you like to do? Choose one of the above options: 7
```

```
Search by:
1. Student name
2. Student ID
3. Student email
4. Back
```

```
Choose: 1
Enter search value: Ethan
```

Search results for 'Ethan' in MAT202B:

Student Id	Name	Grade	Marks
S004	Ethan Williams	C	72
S041	Ethan Garcia	D	67
S054	Ethan Thomas	D	60
S076	Ethan Gonzalez	A	91

4 result(s) found in 0.000121 seconds.

***** Professor Menu *****

```
1. View my profile
2. Display course grade report
3. Manage student grade(s)
4. View course details
5. Add student(s) to my course
6. Remove student from my course
7. Search student records
8. Change password
9. Logout
```

What would you like to do? Choose one of the above options: 7

```
Search by:
1. Student name
2. Student ID
3. Student email
4. Back
```

```
Choose: 2
Enter search value: S003
```

Search results for 'S003' in MAT202B:

Student Id	Name	Grade	Marks
S003	Mia Davison	F	59

1 result(s) found in 0.000033 seconds.

```
Search by:
1. Student name
2. Student ID
3. Student email
4. Back
Choose: 3
Enter search value: mia.davis03@university.edu
```

Search results for 'mia.davis03@university.edu' in MAT202B:

Student Id	Name	Grade	Marks
S003	Mia Davison	F	59

1 result(s) found in 0.000305 seconds.

- Includes sort functionality in Course details to display data based on first/last name
- In course grade report there is additional sort by grade, or mark

```

=====
Total students enrolled: 35
Would you like to sort the records? (yes/no): yes

Sort by:
1. Name (Last name, then First name)
2. Marks
3. Grade
Choose: 2

Order:
1. Ascending
2. Descending
Choose: 2

Student Id      Name                Grade  Marks
-----
S009      Lucas Garcia        A      99
S003      Mia Davison         A      98
S010      Benjamin Davis      A      98
S044      Sophia Hernandez    A      98
S078      Olivia Miller       A      96
S085      Mia Johnson         A      95
S089      Isabella Williams   A      95
S046      James Taylor        A      94
S056      Ava Johnson         A      93
S067      Isabella Moore       A      93
S091      Emma Davis          A      91
S074      Liam Williams       B      89
S092      Isabella Anderson   B      87
S031      Ava Jackson         B      86
S039      Abigail Miller      B      83
S071      Lucas Taylor        B      82
S064      Isabella Smith       C      79
S017      Isabella Johnson    C      78
S033      Olivia Martin       C      78
S082      Abigail Lopez       C      78
S027      Ava Brown           C      76
S032      Logan Jackson       C      75
S077      James Moore         C      74
S004      Ethan Williams      C      70
S087      Harper Jones        D      68
S075      Evelyn Martin       D      67
S016      Evelyn Martin       D      65
S001      UpdatedName Jones   D      64
S036      Mia Jones           D      63
S057      Benjamin Gonzalez   D      62
S083      Harper Garcia       D      62
S086      Isabella Anderson   D      61
S070      Harper Martinez     F      57
S023      Ethan Garcia        F      53
S096      Jane Doe            0

```

Student Profile

- Students can view their profile, containing personal information, course enrollments, and grades

```

What would you like to do? Choose one of the above options: 4
===== Student Profile =====
Student ID: S003
Name:      Mia Davis
Email:     mia.davis03@university.edu
=====

Enrolled Courses:
Course ID   Course Name                Credits
-----
HIS330F     World History              3
CHE115G     General Chemistry          3
CSE101A     Introduction to Programming 6
PHY210D     N/A                        N/A
MAT202B     Calculus II                4

```

Generate grade report by student

```
What would you like to do? Choose one of the above options: 4

===== Grade Report: Mia Davis =====
Course ID   Course Name           Credits   Grade   Marks
-----
HIS330F     World History          3         A       98
CHE115G     General Chemistry      3         F       50
CSE101A     Introduction to Programming 6         D       65
PHY210D     N/A                   N/A        D       60
MAT202B     Calculus II            4         F       59
=====

Average mark: 66.40
Median mark: 60.00

Grade Distribution:
Grade   Count   Percentage
-----
A       1       20.0%
B       0       0.0%
C       0       0.0%
D       2       40.0%
F       2       40.0%
=====
```

- Students can view just Marks/Grades for each class specifically

```
***** Student Menu *****
1. Check my grades
2. Check my marks
3. View my profile
4. View grade report
5. View course catalog
6. Change password
7. Logout
What would you like to do? Choose one of the above options: 1
Mia Davis's Grades
Course ID   Course Name           Grade
-----
HIS330F     World History          A
CHE115G     General Chemistry      F
CSE101A     Introduction to Programming D
PHY210D     N/A                   D
MAT202B     Calculus II            F
***** Student Menu *****
1. Check my grades
2. Check my marks
3. View my profile
4. View grade report
5. View course catalog
6. Change password
7. Logout
What would you like to do? Choose one of the above options: 2
Mia Davis's Marks
Course ID   Course Name           Marks
-----
HIS330F     World History          98
CHE115G     General Chemistry      50
CSE101A     Introduction to Programming 65
PHY210D     N/A                   60
MAT202B     Calculus II            59
```

Unit Tests

- Test student addition/deletion/modification. Perform test to have 1000 records in student file

```
[pass] Sort by marks descending in 0.000167 seconds. Range: 100 -> 0
ok
test_add_student (__main__.TestStudentRecords.test_add_student) ...
[pass] Student added. Count: 1000 -> 1001
ok
test_delete_student (__main__.TestStudentRecords.test_delete_student) ...
[pass] Student deleted. Count: 1000 -> 999
ok
test_has_1000_records (__main__.TestStudentRecords.test_has_1000_records) ...
[pass] Total student records: 1000
ok
test_load_from_csv (__main__.TestStudentRecords.test_load_from_csv) ...
[pass] Loaded 1000 records from test_students.csv successfully.
ok
test_modify_student (__main__.TestStudentRecords.test_modify_student) ... Profile updated successfully.
[pass] Student modified. New first name: UpdatedName
ok
```

- Load data saved in csv files and search. Print total time taken in search cases. Sort student records based on marks or student email address. Include timing it took to sort the records

```
ok
test_search_by_email (__main__.TestSearch.test_search_by_email) ...
[pass] Search by email 'harper.jones01@university.edu': 5 result(s) found in 0.000142 seconds
ok
test_search_by_name (__main__.TestSearch.test_search_by_name) ...
[pass] Search by name 'Jones': 49 result(s) found in 0.000165 seconds
ok
test_search_by_student_id (__main__.TestSearch.test_search_by_student_id) ...
[pass] Search by ID 'S001': 5 result(s) found in 0.000036 seconds
ok
test_search_no_results (__main__.TestSearch.test_search_no_results) ...
[pass] Search no results: 0 found in 0.000138 seconds
ok
test_sort_by_email_ascending (__main__.TestSort.test_sort_by_email_ascending) ...
[pass] Sort by email ascending in 0.000361 seconds.
ok
test_sort_by_email_descending (__main__.TestSort.test_sort_by_email_descending) ...
[pass] Sort by email descending in 0.000355 seconds.
ok
test_sort_by_marks_ascending (__main__.TestSort.test_sort_by_marks_ascending) ...
[pass] Sort by marks ascending in 0.000259 seconds. Range: 0 -> 100
ok
test_sort_by_marks_descending (__main__.TestSort.test_sort_by_marks_descending) ...
[pass] Sort by marks descending in 0.000167 seconds. Range: 100 -> 0
ok
```

- Add/delete/modify the course

```
[pass] C:\Users\amj\OneDrive\Desktop\AI - 2\Lab 1 & 2\code\02m\python3\709019\amj\amj\Desktop\Lab 1\test_run_02m.py
test_add_course (__main__.TestCourseRecords.test_add_course) ... Course added successfully.
[pass] Course added: BI0110 - Biology (3 credits)
ok
test_add_duplicate_course (__main__.TestCourseRecords.test_add_duplicate_course) ... Course ID already exists.
[pass] Duplicate course rejected.
ok
test_delete_course (__main__.TestCourseRecords.test_delete_course) ... Course deleted successfully.
[pass] Course deleted: CS101
ok
test_delete_nonexistent_course (__main__.TestCourseRecords.test_delete_nonexistent_course) ... Course not found.
[pass] Delete nonexistent course rejected.
ok
test_modify_course_credits (__main__.TestCourseRecords.test_modify_course_credits) ... Course updated successfully.
[pass] Course credits modified: DATA200 -> 4 credits
ok
test_modify_course_name (__main__.TestCourseRecords.test_modify_course_name) ... Course updated successfully.
[pass] Course name modified: DATA200 -> Advanced Data Science
ok
test_modify_course_name_and_credits (__main__.TestCourseRecords.test_modify_course_name_and_credits) ... Course updated successfully.
[pass] Course modified: CS101 -> Advanced CS (3 credits)
ok
```

- Add/delete/modify professors

```
OK
test_add_duplicate_professor (__main__.TestProfessorRecords.test_add_duplicate_professor) ... Professor ID already exists.
[pass] Duplicate professor rejected.
OK
test_add_professor (__main__.TestProfessorRecords.test_add_professor) ... Professor added successfully.
[pass] Professor added: P003 - Jane Doe
OK
test_delete_nonexistent_professor (__main__.TestProfessorRecords.test_delete_nonexistent_professor) ... Professor not found.
[pass] Delete nonexistent professor rejected.
OK
test_delete_professor (__main__.TestProfessorRecords.test_delete_professor) ... Professor deleted successfully.
[pass] Professor deleted: P002
OK
test_modify_professor_course (__main__.TestProfessorRecords.test_modify_professor_course) ... Professor details updated successfully.
[pass] Professor course modified: P002 -> DATA200
OK
test_modify_professor_rank (__main__.TestProfessorRecords.test_modify_professor_rank) ... Professor details updated successfully.
[pass] Professor rank modified: P001 -> Distinguished Professor
OK
```


